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TCMSTD 1.0: a systematic analysis of the traditional Chinese medicine system toxicology database

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Dear Editor,

The safety and modernization of traditional Chinese medicine (TCM) are significant concerns for users' lives and health. Recently, the adverse effects caused by TCM and traditional nontoxic TCM have been frequently reported (Joo et al., 2020; Liu et al., 2021). As the safety of TCM continues to attract widespread attention, many researchers have been investigating the toxicity of TCM in regard to the toxic ingredients, effects, mechanisms, and detoxification strategies in TCM (Figure S1 in Supporting Information). Although some TCMs cause toxic reactions, they also play key roles in treating certain unignorable diseases (Fang et al., 2022). Therefore, toxicity research is crucial and urgently needed to develop and safely use TCM.

Currently, database technology efficiently and conveniently aids researchers in mining, storing, searching, and analyzing the information on TCMs. The use of database technology to systematically integrate and analyze the research results of TCM will promote and help clinical research on TCM (Luo et al., 2022). However, the pharmacological effects of TCMs and their constituents are the main emphases of existing TCM databases (Fang et al., 2021; Ru et al., 2014; Xu et al., 2019). Moreover, the current databases do not distinguish between toxic and pharmacodynamic effects on targets, classify and summarize targets for specific toxic manifestations, and support evidence from previous studies (Davis et al., 2021; Lim et al., 2010). Considering the aforementioned issues, we constructed the Traditional Chinese Medicine System Toxicology Database (TCMSTD; https://www.bic.ac.cn/TCMSTD/), which is the first database locally and internationally that focuses on the safe use of TCMs. This study aimed to systematically integrate and analyze the research results of toxic TCMs and provide the functions of the toxic effect and toxic target prediction of TCM ingredients to enable TCM toxicity research.

The construction of TCMSTD was divided into two main stages. The first stage involved data collection and processing (Figure 1A). We attempted to extract the literature content using computer technology. However, we determined that, because of the varied literature format and content writing manner, the computer information extraction method will cause information omissions and extraction errors. Thus, we used the manual data collection method. The inclusion criteria and finalization modules for TCMSTD (Table S1 in Supporting Information) are as follows: (i) toxic TCMs (252 species): we summarized 9,139 TCMs from 10 authoritative works on TCMs (Table S2 in Supporting Information) and screened out the TCMs with clinical poisoning reports or proven toxicity through toxicological experiments, consequently including 252 kinds of toxic

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TCMs in the database. To elaborately identify toxic TCMs, we collected 150 TCMs from the Teaching and Research Section of Traditional Chinese Medicine Resources of Tianjin University of Traditional Chinese Medicine and Pharmacy of Baokang Hospital and photographed them according to the characteristics of TCM. (ii) Formulas (22 species): formulas excluding those of injections and sanitizers were collected according to the Adverse Drug Reaction Information Bulletin issued by the National Medical Products Administration (China). (iii) Chemical ingredients (4,361 species): TCM ingredients were primarily collected based on review articles published over the last five years, and volatile ingredients were excluded as much as possible. (iv) Toxic ingredients (226 species): in addition to meeting the standards of Item (iii), only the ingredients confirmed by toxicological experiments will be considered toxic ingredients. (v) Toxic targets (2,425 species): we systematically classified and summarized five toxic targets according to the most significant toxicity manifestations caused by TCMs, namely, cardiotoxicity, hepatotoxicity, nephrotoxicity, neurotoxicity, and reproductive toxicity (Figure S2A in Supporting Information), constituting the toxic target sub-database of TCMSTD. The aforementioned contents refer to databases, including TCMSP (Ru et al., 2014), PubChem (Kim et al., 2021), ChemSpider (Pence and Williams, 2010), NCBI (Sayers et al., 2021), UniProt (UniProt, 2021), OMIM (Hamosh et al., 2021), and DisGeNET (Piñero et al., 2021). The data are mainly obtained from five literature websites, including CNKI, VIP, WanFang, PubMed, and Web of Science. A total of 5,522 valid articles were included. The aforementioned five modules are interconnected (Figure S2B in Supporting Information). Figure S3 in Supporting Information shows the specific data types that are incorporated in TCMSTD.

The second stage of database construction involved designing and building the modules, functions, and interfaces of the database according to the data collected in the first stage (Figure 1B). TCMSTD was implemented as a web application using JavaScript and HTML for front-end development. The core JavaScript libraries used include Vue.js (https://vuejs.org/) as the main front-end framework, plotly.js (https://plotly.com/) for Lollipop chart creation, and vis.js (https://visjs.org) for network display. The high-level web framework Django (https://www.djangoproject.com/) was used for back-end data preprocessing and analysis. The open-source data management system MySQL was used for table data saving and accessing.

The TCMSTD has Chinese and English interfaces to set up five functions (Figure S4 in Supporting Information). These five functions are as follows: (i) browse function: TCMSTD has browsing interfaces in Chinese and English, and the user can click the "Browse" button on the home page of the database or directly click the target module at the bottom of the home page. (ii) Search function: the global search function is based on the Elasticsearch module. TCMSTD provides search options for five elements (i.e., TCM, formula, chemical ingredient, toxic ingredient, and toxic target), and the user can see the corresponding keyword input in the search box at the upper right corner of the home page of the database and proceed to the details page, thus realizing the function. (iii) Predictive function: two sub-databases are embedded in TCMSTD, namely, the toxic ingredient and toxic target databases. They can realize two functions by predicting whether the ingredient is toxic or not and predicting the toxic target of a toxic ingredient. This function uses the Tanimoto coefficient based on the Open Babel package to evaluate the similarity of chemical ingredients. By default, query chemicals that have a Tanimoto coefficient greater than 0.6 and one molecule in TCMSTD would be treated as having similar toxic targets. Users can click the "Tools" option on the home page and enter the ingredient structure in the blank space to use this function. The results page will show the list of results and network diagram. Table S3 in Supporting Information shows the list of toxic target results of predicted toxic ingredients. (iv) Systematic analysis function: the "System Analysis Function" option is displayed on the home page of the database, and it is available upon click. The required elements (i.e., TCM, formula, ingredient, and toxic target) are selected, and their network relationships are built based on the network module of the visualization database vis.js (https://visjs.org). Pathway enrichment analysis of the Gene Ontology and Reactome pathway information is conducted. The enriched pathway is computed using a hypergeometric test to derive the significant p-value, which is further corrected using the Benjamini and Hochberg algorithm. (v) Download function: the TCMSTD can download data tables and predictive analysis results. Each part of the downloadable content is provided with text or icon prompts. Even without user registration or login, the aforementioned five features can be enabled quickly, conveniently, and freely. Figure S5 in Supporting Information shows the application case of TCMSTD.

Data update is an extremely important task of the database, which aims to ensure a timely display of the current research progress on TCM and reflect the significance of the existence of the database. Every six months, TCMSTD will be updated, and new versions will be released on June 30 and December 30 yearly. We always pay attention to the research progress on toxic TCMs to promptly update the invalid data in the database and supplement the missing data and the latest research results. In subsequent updated versions, we will update and upgrade the algorithm and program according to the development of computers and the actual work needs of TCM toxicity researchers. As a development team, we have administrative rights to the database background; thus, close attention will be paid to the database status at any

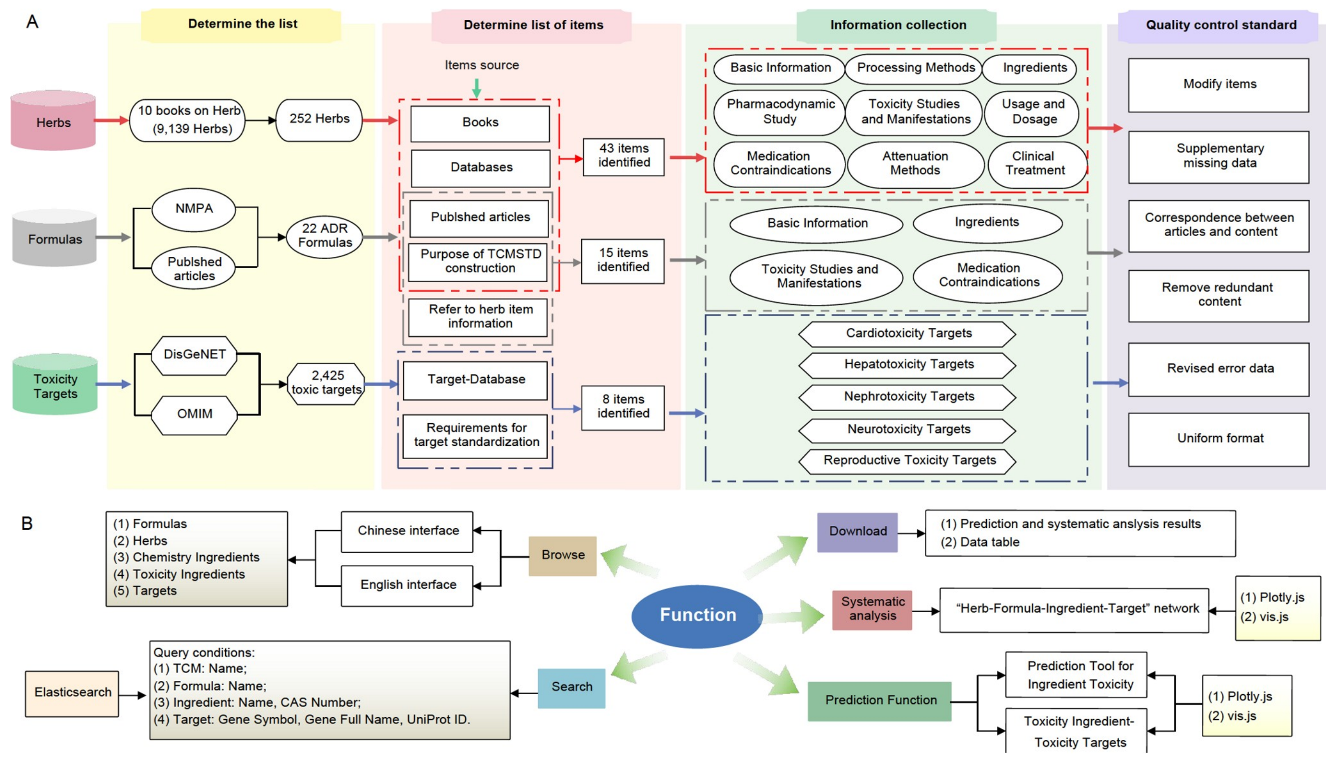

Figure 1 Database construction process: (A) data preparation process and (B) functional construction of Traditional Chinese Medicine System Toxicology Database.

time to identify problems and respond appropriately, ensuring problem-free utilization of the database by users.

In summary, TCMSTD, the first database focusing on TCM safety, has three innovations: first, mining and system integration and analysis of toxic-TCM information; second, the toxic target sub-database covering five toxic effects embedded in TCMSTD; and third, the use of TCMSTD to predict toxicity based on the structural information of ingredients. A qualified database should have a reasonable data structure, a small degree of data redundancy, and accurate and reliable results included in the information and analysis system, and can track the research status of toxic-TCM in real-time to make timely integration and analysis, which is the purpose of the construction of the TCMSTD. We are confident that TCMSTD will be an effective tool for promoting TCM safety usage and providing an efficient and comprehensive platform to researchers engaged in TCM toxicity research, TCM safety regulators, and drug users for sharing information on TCM toxicology.

Compliance and ethics *The author(s) declare that they have no conflict of interest.*

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References

Davis, A.P., Grondin, C.J., Johnson, R.J., Sciaky, D., Wiegers, J., Wiegers, T.C., and Mattingly, C.J. (2021). Comparative Toxicogenomics Database (CTD): update 2021. Nucleic Acids Res 49, D1138–D1143.

Fang, S.S., Dong, L., Liu, L., Guo, J.C., Zhao, L.H., Zhang, J.Y., Bu, D.C., Liu, X.K., Huo, P.P., Cao, W.C., et al. (2021). HERB: a high-throughput experiment- and reference-guided database of traditional Chinese medicine. Nucleic Acids Res 49, D1197–D1206.

Fang, Z., Guo, Y., Wang, Z., He, T., Wang, J., Bai, Z., and Xiao, X. (2022). Asari Radix et Rhizoma consumption lacks relevance for hepatocellular carcinoma in patients: a retrospective cohort study. Chin Herb Med 14, 470–475.

Hamosh, A., Amberger, J.S., Bocchini, C., Scott, A.F., and Rasmussen, S. A. (2021). Online Mendelian Inheritance in Man (OMIM®): Victor McKusick's magnum opus. Am J Med Genet A 185, 3259–3265.

Joo, Y.J., Newcombe, D., Nosa, V., and Bullen, C. (2020). Investigating betel nut use, antecedents and consequences: a review of literature. Subst Use Misuse 55, 1422–1442.

Kim, S., Chen, J., Cheng, T., Gindulyte, A., He, J., He, S., Li, Q., Shoemaker, B.A., Thiessen, P.A., Yu, B., et al. (2021). PubChem in 2021: new data content and improved web interfaces. Nucleic Acids Res 49, D1388–D1395.

Lim, E., Pon, A., Djoumbou, Y., Knox, C., Shrivastava, S., Guo, A.C., Neveu, V., and Wishart, D.S. (2010). T3DB: a comprehensively annotated database of common toxins and their targets. Nucleic Acids Res 38, D781–D786.

Liu, X., Li, H., Wang, S., Zhang, J., and Liu, D. (2021). Sesquiterpene lactones of *Aucklandia lappa*: pharmacology, pharmacokinetics, toxicity, and structure-activity relationship. Chin Herb Med 13, 167–176.

Luo, L., Yang, J., Wang, C., Wu, J., Li, Y., Zhang, X., Li, H., Zhang, H., Zhou, Y., Lu, A., et al. (2022). Natural products for infectious microbes and diseases: an overview of sources, compounds, and chemical diversities. Sci China Life Sci 65, 1123–1145.

Pence, H.E., and Williams, A. (2010). ChemSpider: an online chemical information resource. J Chem Educ 87, 1123–1124.

Piñero, J., Saüch, J., Sanz, F., and Furlong, L.I. (2021). The DisGeNET cytoscape app: exploring and visualizing disease genomics data. Comput Struct Biotechnol J 19, 2960–2967.

Ru, J., Li, P., Wang, J., Zhou, W., Li, B., Huang, C., Li, P., Guo, Z., Tao, W., Yang, Y., et al. (2014). TCMSP: a database of systems pharmacology for drug discovery from herbal medicines. J Cheminform 6, 13.

Sayers, E.W., Beck, J., Bolton, E.E., Bourexis, D., Brister, J.R., Canese, K., Comeau, D.C., Funk, K., Kim, S., Klimke, W., et al. (2021). Database resources of the National Center for Biotechnology Information. Nucleic Acids Res 49, D10–D17.

UniProt, C. (2021). UniProt: the universal protein knowledgebase in 2021. Nucleic Acids Res 49, D480–D489.

Xu, H.Y., Zhang, Y.Q., Liu, Z.M., Chen, T., Lv, C.Y., Tang, S.H., Zhang, X. B., Zhang, W., Li, Z.Y., Zhou, R.R., et al. (2019). ETCM: an encyclopaedia of traditional Chinese medicine. Nucleic Acids Res 47, D976–D982.

SUPPORTING INFORMATION

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